AI-BASED SALES PREDICTION AND FORECASTING USING MACHINE LEARNING

A Comparative Analysis of Random Forest, Gradient Boosting, XGBoost and Time-Series Forecasting Using Walmart Sales Data

ACADEMIC PROJECT REPORT

Prepared from the supplied project README and verified experiment summaries

Project Domain: Machine Learning | Retail Sales Analytics
Dataset: Walmart Historical Weekly Sales | 45 Stores | 6,435 Observations

Project Overview

What the project does: This project analyses historical Walmart weekly sales and uses machine learning to estimate future retail sales. It compares predictive models, examines how forecasting errors vary across stores and periods, and evaluates the reliability of prediction intervals and sales-movement warnings.

In practical terms, the system uses previous sales patterns to estimate how much each store may sell in a future week. Such estimates can support demand understanding, inventory planning and business operations. The report also makes clear that historical model performance does not guarantee future accuracy.

Abstract

Sales forecasting is an important part of retail business management. Accurate sales predictions can help organizations understand customer demand, manage inventory, plan resources and make informed business decisions. This project, titled **AI-Based Sales Prediction and Forecasting Using Machine Learning**, develops and evaluates forecasting models using historical Walmart sales data.

The dataset contains 6,435 observations from 45 retail stores, covering 5 February 2010 through 26 October 2012. Random Forest, Gradient Boosting and XGBoost are compared with Last-Week and Seasonal baseline methods. Historical sales lags, rolling summaries and calendar-related information are used in the forecasting workflow.

Performance is evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Weighted Absolute Percentage Error (WAPE) and R-squared (R^2). Further analyses examine nested backtesting, store-level errors, prediction intervals, feature ablation and warning usefulness. The results show that model performance varies across periods: Random Forest performs strongly in the original benchmark, while the Seasonal Baseline wins the pooled nested backtest. The project therefore emphasizes careful comparison and honest interpretation of forecasting uncertainty.

1. Introduction

Retail sales vary over time and across stores. Holidays, seasonal patterns and historical demand can influence weekly sales. Businesses use forecasts to anticipate demand and make operational decisions. Machine learning offers methods for learning patterns from historical observations, but its performance must be assessed against simpler forecasting approaches.

This project develops a forecasting workflow using Walmart weekly sales data. It evaluates three machine learning families and two baselines, then investigates how prediction errors, feature contributions and warning behavior change across stores and time periods.

2. Problem Statement

Retail businesses need reliable estimates of future sales to manage inventory, understand demand and plan operations. Forecasting is challenging because sales patterns differ across stores and time periods, and may be affected by seasonal changes, holidays and broader conditions. The project addresses the problem of developing and evaluating models that predict weekly retail sales from historical Walmart sales information while examining the reliability and limitations of those predictions.

3. Objectives

- Analyse historical Walmart weekly sales across multiple stores.
- Develop forecasting models using Random Forest, Gradient Boosting and XGBoost.
- Compare machine learning models with Last-Week and Seasonal baselines.

- Evaluate predictions using MAE, RMSE, WAPE and R^2.
- Examine historical sales features, feature importance and feature ablation.
- Assess store-level prediction errors and prediction-interval coverage.
- Evaluate sales-movement warnings using precision, recall and missed/false alerts.
- Generate forecasts using the selected forecasting configuration.

4. Dataset Description

The project uses historical Walmart retail sales data. It contains 6,435 rows, 45 stores and 143 consecutive weekly dates per store, from 5 February 2010 to 26 October 2012. The supplied project documentation reports no missing cells or duplicate store-date rows.

Attribute	Description
Store	Store identification number.
Date	Weekly sales date.
Weekly_Sales	Sales recorded for the store during that week.
Holiday_Flag	Indicates whether the week contains a holiday.
Temperature	Temperature information.
Fuel_Price	Fuel price information.
CPI	Consumer Price Index.
Unemployment	Unemployment rate.

Sales are treated as USD by convention in the project metrics, but the original source currency metadata remains unverified. Same-week weather and economic predictors are excluded because their availability at the forecast origin is not verified.

5. Methodology

5.1 Data preparation

The data is checked for missing values, duplicate store-date records, date consistency, numeric validity and panel alignment. Observations are kept in chronological order to prevent future information from leaking into historical predictions.

5.2 Feature engineering

The workflow creates store-wise, past-only lag features and shifted rolling summaries. It includes previous sales, longer historical lags, prior-year summaries and maintained calendar features. A 53-week warm-up leaves 4,050 supervised rows: 3,420 development rows and 630 test rows.

5.3 Forecast design

The final 14 dates, 27 July to 26 October 2012, are held fixed for the original test evaluation. The evaluation is one-week-ahead: predictions can use finalized actual sales through the preceding week, including earlier dates in an evaluation block, while fitted models remain fixed within each block. This is not an all-at-once multiweek forecast.

5.4 Machine learning algorithms

- **Random Forest:** An ensemble of decision trees whose predictions are combined.
- **Gradient Boosting:** Builds sequential models that attempt to reduce the errors of earlier models.
- **XGBoost:** An optimized gradient-boosting implementation used for predictive modelling.

5.5 Baseline methods

Two simpler methods provide reference points: the Last-Week Baseline uses the previous week's sales, while the Seasonal Baseline uses historical seasonal patterns.

5.6 Evaluation metrics

- **MAE:** Average absolute difference between actual and predicted sales.
- **RMSE:** Square root of mean squared error; larger errors receive greater weight.
- **WAPE:** Total absolute error divided by total actual sales, expressed as a percentage. WAPE is not accuracy.
- **R^2:** A secondary measure describing variation explained by a model.

6. Experimental Results

6.1 Original benchmark comparison

The following table preserves the original benchmark metrics. These are not the scores of the revised unified forecast workflow.

Model	MAE	RMSE	WAPE	R^2
Random Forest	36,036.48	56,558.73	3.51%	0.9884
Gradient Boosting	36,167.74	53,824.53	3.53%	0.9895
XGBoost	40,119.75	59,765.80	3.91%	0.9871
Last-week baseline	48,895.62	73,367.26	4.77%	0.9805
Seasonal baseline	52,325.36	83,959.12	5.10%	0.9745

Random Forest has the lowest original benchmark MAE and WAPE, while Gradient Boosting has the lowest RMSE. The Random Forest and Gradient Boosting MAE difference is only $131.26, and each wins 7 of the 14 test weeks. The available uncertainty interval for their MAE difference includes zero, so the project does not claim universal superiority.

6.2 Nested backtesting

Four non-overlapping development blocks cover 5 August 2011 to 20 July 2012. At each origin, model configurations are tuned using earlier folds. All five methods predict identical rows, totaling 51 weeks and 2,295 store-weeks per method.

Model	Pooled MAE	Pooled RMSE	WAPE
Seasonal baseline	61,207.48	88,498.27	5.74%
Random Forest	69,422.13	141,344.31	6.52%
Gradient Boosting	70,489.50	137,922.15	6.62%
XGBoost	74,863.54	147,567.81	7.03%

Model	Pooled MAE	Pooled RMSE	WAPE
Last-week baseline	98,545.96	191,208.72	9.25%

The Seasonal Baseline wins pooled backtest MAE and the 2011 shopping peak. This additional evidence challenges a general claim that machine learning is superior throughout the year. These development diagnostics are not a second untouched final test.

7. Forecast Reliability Analysis

The earlier frozen-model prediction band had 94.44% aggregate coverage but concealed substantial store-level failures. The revised unified point-and-band workflow achieved 98.89% overall empirical coverage on the reviewed holdout.

Workflow	Overall coverage	Store 14	Store 36
Replaced frozen-model band	94.44%	7.14%	50.00%
Unified point/band workflow	98.89%	92.86%	78.57%

One store, Store 36, remains below the nominal 90% coverage level in the updated workflow. The revised mean interval width is $340,857.45, compared with $210,666.28 previously. Wider intervals improve observed coverage but reduce planning precision. These empirical residual-transfer bands do not guarantee future coverage for every store.

8. Feature Importance and Ablation

Feature ablation removes sales_lag_51, sales_lag_52, sales_lag_53 and prior_year_mean_3 while retaining the original selected model parameters and evaluation rows. The reported average MAE changes after removing these features are:

Model	All-fold mean MAE change	First-fold share of net increase	Later-fold mean change
Gradient Boosting	130.66%	100.62%	-1.42%
Random Forest	120.47%	98.57%	3.10%
XGBoost	82.25%	98.78%	1.72%

The first evaluation block accounts for most of the mean performance loss. Later effects are much smaller and sometimes reversed. This suggests that the contribution of historical sales features varies by period; the results are not causal claims or evidence of a uniform annual-lag benefit.

9. Sales Movement and Warning Analysis

The project evaluates whether warning rules identify material sales movements. A material movement is defined as an actual change of at least ±15% from the previous 13-week mean. Thresholds are estimated using information available strictly before each backtest origin.

Warning rule	Alerts	Material moves	Precision	Recall	False alarms	Missed moves
Always NORMAL	0	396	N/A	0.00%	0	396
Fixed ±15%	181	396	84.53%	38.64%	28	243

Warning rule	Alerts	Material moves	Precision	Recall	False alarms	Missed moves
Origin quantiles	97	396	91.75%	22.47%	8	307

The origin-quantile method has higher precision and fewer false alarms, while the fixed-threshold method captures more material movements. The results demonstrate a trade-off between alert volume, precision and missed movements. They do not establish business profitability or anomaly probabilities.

10. Final Forecasting Configuration

The revised unified forecasting workflow uses Random Forest with maximum depth set to None and minimum samples per leaf set to 1. The configuration is selected using inner validation before 6 April 2012. The same learner and settings are then refitted on all development rows for holdout prediction and on all available labelled rows for the unseen forecast.

Metric	Revised workflow result
MAE	$35,910.00
RMSE	$55,967.12
WAPE	3.50%
Empirical band coverage	98.89%
Mean band width	$340,857.45
Forecast date	2 November 2012

The revised holdout results must be distinguished from the original benchmark. The forecast is for 2 November 2012, immediately after the historical file ends. No realized accuracy, warning utility or future store coverage is available for that date.

11. Advantages

- Compares several machine learning algorithms and traditional baselines.
- Uses historical sales patterns across 45 stores.
- Includes chronological evaluation and past-only feature construction.
- Reports multiple performance metrics rather than relying on one score.
- Examines individual-store errors and prediction interval coverage.
- Evaluates warning precision, recall, false alarms and missed movements.
- Provides a reproducible notebook-based workflow with verification checks.

12. Limitations

- The dataset covers only February 2010 through October 2012.
- Only one Thanksgiving/Christmas cycle is testable after the historical warm-up.
- The future forecast has no corresponding actual sales in the supplied dataset.
- Prediction intervals do not guarantee future or individual-store coverage.

- Same-week weather and economic predictor availability is unverified, so these fields are excluded from forecasting features.
- The dataset's source currency, exact provenance and redistribution permission remain unverified.
- The revised workflow was designed after reviewing holdout failures; its revised results are retrospective diagnostics and require independent future validation.

13. Future Scope

- Use more recent Walmart or other retail sales observations.
- Evaluate additional machine learning and deep learning forecasting methods.
- Include multiple annual cycles to better assess seasonal behavior.
- Develop an interactive sales forecasting dashboard.
- Integrate new sales observations and evaluate forecasts prospectively.
- Improve store-specific prediction interval reliability.
- Explore inventory planning and demand-management applications.

14. Conclusion

This project demonstrates a complete machine learning workflow for retail sales prediction using historical Walmart sales data. Random Forest, Gradient Boosting and XGBoost are compared with Last-Week and Seasonal baselines. In the original benchmark, Random Forest achieves the lowest MAE and WAPE, while Gradient Boosting achieves the lowest RMSE. In nested backtesting, however, the Seasonal Baseline achieves the lowest pooled MAE.

The additional analyses show that forecasting performance varies across time periods and stores. Prediction interval coverage, feature contributions and warning usefulness involve trade-offs that should be considered before operational use. The project therefore avoids claiming that one model is universally best or that historical performance guarantees future accuracy.

Overall, the project delivers a reproducible machine learning-based sales forecasting workflow that analyses historical retail sales, compares predictive models, evaluates reliability and generates historical-data-based future sales estimates.

Appendix A. Running the Project

The supplied README describes a local VS Code workflow. Open the project folder, enable the Python and Jupyter extensions, select the project's .venv Python environment, restart the kernel and run all notebook cells in order. The final verification section should print a PASS status. The notebook and data file are required to reproduce the computational results.

The documented terminal alternative, from the project root with the environment activated, is:

```
python -m jupyter nbconvert --to notebook --execute --inplace --ExecutePreprocessor.timeout=600 Sales_Forecasting_ML.ipynb
```

The README reports successful verification of 27 code cells, 19 embedded figures, numerical checks and the complete report in the tested environment. Cross-platform execution beyond the stated verification environment has not been confirmed.